

Erik Jensko

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Keywords	Theoretical cosmology; interacting dark sectors; effective field theory; cosmological stability; dynamical systems; modified gravity; CLASS/Cobaya and Bayesian cosmology.	
Research Experience	University College London Department of Mathematics <i>EPSRC Doctoral Prize Postdoctoral Research Fellow</i>	2024 - 2026
Education	University College London <i>Ph.D. Mathematics</i> (awarded December 28th 2023) Thesis: ‘A unified approach to geometric modifications of gravity’ Advisor: Prof. Christian G. Boehmer	2019 - 2023
	University of Nottingham <i>M.Sc. Gravity, Particles and Fields (Distinction)</i> Thesis: ‘Scalar fields in cosmology: a dynamical systems approach’ Advisor: Prof. Edmund J. Copeland	2017 - 2018
	University of Nottingham <i>B.Sc. Physics with Theoretical Physics (Hons)</i> Thesis: ‘Vacuum instability of the Standard Model of Particle Physics’ Advisor: Dr. Peter Millington	2014 - 2017
Publications	12. <i>Non-minimal fluid Lagrangian couplings</i> C. G. Boehmer, E. Jensko[†], E. Al-Nasrallah Phys.Rev.D 114 (2026) 4, 044080 [2605.16220 gr-qc] 11. <i>Interacting dark sector from intrinsic entropy couplings</i> E. Jensko[†], E. M. Teixeira and V. Poulin Phys.Rev.D 114 (2026) 2, 023536 [2603.10622 astro-ph.CO] 10. <i>Cosmological dynamical systems of non-minimally coupled fluids and scalar fields</i> H. Ashi, C. Boehmer, A. del Sordo, E. Jensko[†] Gen.Rel.Grav. 57 (2025) 12, 167 [2509.14440 gr-qc] 9. <i>Geometric formulation of k-essence and late-time acceleration</i> L. Csillag and E. Jensko[†] JCAP 09 (2026) 028 [2505.15975 gr-qc] 8. <i>The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics</i> (contributor) Phys.Dark Univ. 49 (2025) 101965 [2504.01669 astro-ph.CO] 7. <i>Spatial curvature in coincident gauge $f(Q)$ cosmology</i> E. Jensko[†] Class.Quant.Grav. 42 (2025) 5, 055011 [2407.17568 gr-qc] 6. <i>Relativistic interacting fluids in cosmology</i> D. Iosifidis, E. Jensko and T. Koivisto[†] JCAP 11 (2024) 043 [2406.01412 gr-qc]	

5. *A new 2D formulation of modified General Relativity*
C. Boehmer[†] and **E. Jensko**
Fortsch.Phys. 72 (2024) 6, 2300216 [[2308.01728](#) gr-qc]
4. *Dynamical systems analysis of $f(Q)$ gravity*
C. Boehmer, **E. Jensko** and R. Lazkoz[†]
Universe 9 (2023) 4, 166 [[2303.04463](#) gr-qc]
3. *Modified gravity: A unified approach to metric-affine models*
C. Boehmer and **E. Jensko**[†]
J.Math.Phys. 64 (2023) 8, 082505 [[2301.11051](#) gr-qc]
2. *Cosmological dynamical systems in modified gravity*
C. Boehmer, **E. Jensko** and R. Lazkoz[†]
Eur.Phys.J.C 82 (2022) 6, 500 [[2201.09588](#) gr-qc]
1. *Modified gravity: a unified approach*
C. Boehmer[†] and **E. Jensko**
Phys.Rev.D 104 (2021) 2, 024010 [[2103.15906](#) gr-qc]

Research Funding	<i>UCL Global Engagement Funds (GEF) (Co-applicant)</i>	2026
	<i>Strengthening university research in Bosnia and Herzegovina through mathematical modelling and cosmology</i> University of Tuzla, Bosnia and Herzegovina, July 2026	
	<i>International Visiting Scholar & Remote Collaboration Grant</i>	2023
	<i>Investigating strong coupling with effective field theory methods</i> University of Science and Technology of China, March–May 2023	
Fellowships and Training Grants	<i>EPSRC Doctoral Prize Fellowship extension</i>	2026
	Six-month postdoctoral fellowship extension at UCL, April–October 2026	
	<i>EPSRC Doctoral Prize Fellowship</i>	2024–2026
	Two-year postdoctoral fellowship at UCL, April 2024–April 2026	
	<i>EPSRC PhD Studentship</i>	2019–2023
	<i>Lorentz invariance and diffeomorphism invariance in modified gravity</i> Fully funded PhD studentship, UCL Department of Mathematics	
Travel and Mobility Grants	<i>COST CA24101: Short Term Scientific Mission (STSM)</i>	2026
	<i>Thermodynamic dark sector models and phase-space structures for fundamental physics</i> University of Wrocław, Poland, September 2026	
	<i>UCL MAPS Early Career Researchers Travel Grant</i>	2026
	<i>Interacting dark sector models: cosmology of elastic scattering</i> Universidad Politécnica de Madrid, Spain, June 2026	
	<i>Institute of Physics Early Career Researchers Fund</i>	2025
	Annecy-le-Vieux Laboratory of Theoretical Physics (LAPTh), France, December 2025	
	<i>COST CA21136: Short Term Scientific Mission (STSM)</i>	2025
	<i>Investigating cosmic tensions with generalised interacting dark sector models</i> Universidad de Salamanca, Spain, October 2025	
	<i>COST CA21136: Short Term Scientific Mission (STSM)</i>	2024
	<i>Cosmological dynamics in geometrical gravitational theories</i>	

	Babeş-Bolyai University, Romania, October 2024	
	<i>London Mathematical Society Travel Grant for Early Career Researchers</i>	2023
	<i>Dora Plus Grant for Visiting Doctoral Students</i> University of Tartu, Estonia, August–October 2021	2021
Awards	<i>IOP Publishing Impact Award</i> Selected for a collection celebrating standout UK research for my work on spatial curvature in $f(Q)$ cosmology	2025
	<i>David Warren Fund Scholarship</i> For outstanding work in Graduate Studies at University College London	2022
	<i>Andrew Rosen Prize in Applied Mathematics</i> For outstanding work in Graduate Studies at University College London	2021
Research Visits	University of Science and Technology of China Department of Astronomy / Particle Cosmology Group Host: Prof. Yi-Fu Cai One-month invited research visit, working on extended dark matter models.	November–December 2026
	University of Montpellier Laboratoire Univers et Particules de Montpellier Host: Dr. Vivian Poulin Collaboration on entropy coupled interacting dark sector models.	November 2025
	Universidad de Salamanca IUFFyM / Gravitation and Cosmology Group (GRACUS) Host: Prof. Jose Beltrán Jiménez Collaboration on EFT constructions of dark sector models and perturbative stability.	October 2025
	SISSA Astroparticle and Gravitational Physics Group Host: Prof. Enrico Barausse Collaboration on k -essence and astrophysical screening mechanisms.	June 2025
	Babeş-Bolyai University Department of Physics Host: Prof. Tiberiu Harko Collaboration on geometric modifications of gravity and cosmological constraints.	October 2024
	University of Science and Technology of China Department of Astronomy Host: Prof. Yi-fu Cai Collaboration on effective field theory approaches to strong coupling in modified gravity.	April–May 2023
	University of Tartu Laboratory of Theoretical Physics Host: Prof. Tomi S. Koivisto Collaboration on Lagrangian formulations of relativistic interacting fluids.	August–October 2021
Talks and Presentations	Invited international conference talks	

Lagrangian models of interacting dark matter and dark energy Jul 2026
[GGI Workshop: Exploring New Frontiers in Cosmology](#), Galileo Galilei Institute, Italy

Pure momentum transfer from non-minimal couplings Jun 2026
Mini Workshop on Cosmology with Elastic Dark Interactions: Current Progress and Open Challenges, Universidad Politécnica de Madrid, Spain

Interacting dark sector from entropy couplings Sep 2025
[Tensions in Cosmology, Corfu2025](#), Corfu Summer Institute, Greece

Geometric modifications of gravity Apr 2023
[International Conference of Deep Space Sciences](#), USTC, China

Invited seminar talks

Thermodynamic interpretations of extended dark sector models in cosmology Oct 2026
[Institute of Theoretical Physics Seminar](#), University of Wrocław, Poland

Intrinsic entropy perturbations in cosmology Jun 2026
[CMB/LSS Meeting, Kavli Institute for Cosmology](#), University of Cambridge, UK

Interacting dark sector with intrinsic entropy couplings Mar 2026
[CosmoVerse Journal Club Seminar](#), Online

Interacting dark sector imprints on matter clustering Feb 2026
[High-Energy Physics Seminar](#), University of Southampton, UK

Pure momentum transfer from dark entropy couplings Jan 2026
[London Relativity and Cosmology Seminar](#), Queen Mary University, UK

Intrinsic entropy couplings in the dark sector Oct 2025
[Gravitation and Cosmology Seminar](#), University of Sheffield, UK

Entropy interactions between dark matter and dark energy Oct 2025
[Gravitation and Cosmology Group Seminar](#), Universidad de Salamanca, Spain

Geometric k-essence from nonmetricity and late-time cosmology Jun 2025
[Astroparticle and Gravitational Physics Seminar](#), SISSA, Italy

Interacting fluids in cosmology: Lagrangian approaches and extensions Nov 2024
[Universe and Particles Laboratory Seminar](#), University of Montpellier, France

Advanced dynamical systems methods in cosmology Oct 2024
Physics Department Seminar, Babeş-Bolyai University, Romania

Second-order modifications of gravity Jun 2023
[DAMTP General Relativity Seminar](#), University of Cambridge, UK

Studying $f(T)$ and $f(Q)$ gravity with dynamical systems in cosmology May 2023
[Particle Cosmology Group Meeting](#), USTC, China

Cosmological dynamical systems in modified gravity Jun 2022
[London–Oldenburg Relativity Seminar](#), Online

	<i>Cosmological dynamical systems in modified gravity</i> HEP–Cosmo–Astro Seminar , Online	May 2022
Contributed international conference talks		
	<i>Viable interactions in the dark sector</i> 31st International Symposium on Particles, Strings, and Cosmology (PASCOS) , University of Sheffield, UK	Jun 2026
	<i>Entropy interactions in the dark sector</i> Iberian Cosmology Meetings (IberiCos 2026) , University of Granada, Spain	Apr 2026
	<i>k-essence and cosmic expansion from integrable vectorial nonmetricity</i> 13th Bolyai–Gauss–Lobachevsky Conference on Non-Euclidean Geometry in Modern Physics and Mathematics , Morocco	May 2025
Poster presentations		
	<i>Cosmological fluids in modified gravity</i> COSMO’24 , Kyoto University, Japan	Oct 2024
	<i>Modifying gravity by breaking fundamental symmetries</i> CosmoVerse@Krakow , Jagiellonian University, Poland	Jul 2024
	<i>Metric-affine gravity and its modifications</i> UK Cosmology Meeting , University of Sussex, UK	Jan 2023
Teaching Experience	<i>Teaching</i>	
	<i>Computing for Mathematical Physics</i> (Department of Physics) Senior Postgraduate Teaching Assistant	2022 & 2023
	<i>An Introduction to Mathematica</i> (Department of Mathematics) Postgraduate Teaching Assistant	2021
	<i>UCL Arena Teaching Development Programme</i>	2021–2022
	<i>Supervision</i>	
	- BSc 3rd year project (UCL Department of Mathematics) MAPS Summer Research Internship (UCL Department of Mathematics)	2025 - 2026 2024
Service and Outreach	In2Research Mentor at In2scienceUK	2024 - 2025
	Equality, Diversity and Inclusion (EDI) committee Department of Mathematics, University College London	2021 - present
	Co-organiser of Philosophy and space-time reading group Department of Mathematics, University College London	2024 - present
	Reviewer for <i>Classical and Quantum Gravity</i> , <i>Nuclear Physics B</i> , <i>JCAP</i> , <i>Journal of Mathematical Physics</i> , & <i>Physics Letters B</i>	2023 - present
Computational Expertise	<i>CLASS, Cobaya, Mathematica/xAct, Python, C, Bayesian inference and MCMC methods.</i>	